Active Directory, VirtualBox and Exchange Defensive Lab Roadmap

A practical reference manual for your Windows Server 2022 `josh.lab` lab: what already works, what to build next, which PowerShell commands to use, how to choose VirtualBox networking, how to use the ProBook safely as a remote console, and when to postpone Exchange/OWA.

Prepared for Josh Parris

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Contents

How to use this booklet

This booklet is a practical reference for your Windows Server 2022 Active Directory lab in VirtualBox. It turns our chat into a staged roadmap you can actually follow: first prove AD, DNS, logon, Group Policy and shares; then, only later, consider a separate Exchange/OWA defensive lab.

> **Safety boundary**
>
> This guide does not include exploit payloads, malicious emails, or instructions to reproduce an OWA zero-day. The Exchange/CVE material is limited to defensive validation, logging, mitigation checks, and lab-safe learning.

Recommended reading pattern

1. Read chapters 1-3 before touching the lab.
2. Use chapters 4-8 while building the client VM and joining the domain.
3. Use chapters 9-11 when something breaks.
4. Leave Exchange chapters until the AD foundation is working and you have enough hardware capacity.

Your current lab conclusion

Your server is already functioning as a domain controller for `josh.lab`. ADWS, DNS, KDC and Netlogon were running. DNS resolved `josh.lab`. SYSVOL and NETLOGON were present and online. Exchange Server was not installed: the only `*Exchange*` service was Hyper-V Data Exchange, and `C:\Program Files\Microsoft\Exchange Server\V15` did not exist.

1. Executive summary and current status

Plain English summary

You have successfully built the start of a real Windows domain lab. The next sensible milestone is not Exchange. It is a Windows 10 Pro client VM joined to `josh.lab`, logging in as a test domain user, applying Group Policy, and accessing a domain share.

Area	Current status	Meaning
Windows Server 2022 VM	Working	Server OS is running in VirtualBox.
Active Directory Domain Services	Working	The VM is a domain controller for `josh.lab`.
DNS	Working	The domain name resolves to the DC.
Kerberos and Netlogon	Working	The core domain authentication services are running.
SYSVOL and NETLOGON shares	Online	Group Policy and logon script shares exist.
DFSR event test	Warning	Old setup/startup errors can make dcdiag fail for 24 hours even when service is now running.
Time service	Watch	The VM had big time jumps earlier; AD relies on time accuracy.
Exchange / OWA	Not installed	OWA CVE testing does not apply to this VM yet.
Lab hardware	Constrained	The Lenovo ThinkPad should not be expected to run DC + client + Exchange comfortably all at once.

Best immediate path

1. Keep your current VM as **DC01**.
2. Create a second VM on the ThinkPad called **CLIENT01** using Windows 10 Pro 22H2 x64.
3. Use a VirtualBox Internal Network for AD traffic.
4. Optionally add a second bridged adapter to CLIENT01 so you can remote into it from the ProBook.
5. Join CLIENT01 to `josh.lab`.
6. Test domain login, DNS, Group Policy, and file shares.
7. Only after that, think about Exchange as a separate short-lived or stronger-hardware lab.

2. Safety, ethics, and lab boundary

The lab is for learning Windows Server, Active Directory, domain joins, Group Policy, basic defensive security and safe Exchange mitigation validation. Treat any CVE or OWA work as defensive only.

Allowed in this guide

- Building AD, DNS, users, groups, shares and Group Policy.
- Joining a client VM to a test domain.
- Verifying whether Exchange/OWA exists.
- Checking patch state, mitigation state, service health and logs.
- Learning generic XSS/CSP concepts in toy, non-Exchange examples if needed.

Not included

- No malicious OWA payloads.
- No crafted exploit emails.
- No instructions to bypass MFA, steal sessions, or run script in OWA.
- No testing against DCS, Avance, client systems, or any public IP without written authorisation and scope.

> **Work laptop caution**
>
> Do not join the actual DCS ProBook host to your personal lab domain. If you use the ProBook, use it only as a remote console, and only if you are authorised to use it for lab learning.

3. Lab roadmap

Stage roadmap

Stage	Goal	VMs running	Exit criteria
0. Confirm current DC	Know what already works	DC01 only	ADWS, DNS, KDC, Netlogon running; SYSVOL and NETLOGON online.
1. Clean DC foundations	Static IP, time, DNS health	DC01 only	dcdiag mostly clean or warnings understood; time stable; DC DNS points to itself.
2. Build client VM	Install Windows client	DC01 + CLIENT01	CLIENT01 boots, can ping DC, DNS points to DC.
3. Domain join	Prove core AD stack	DC01 + CLIENT01	CLIENT01 joins `josh.lab`, reboots, `JOSHLAB\testuser` logs in.
4. Group Policy and shares	Practical admin skills	DC01 + CLIENT01	Mapped drive / wallpaper / local admin restriction / test policy applies.
5. Remote console convenience	Use ProBook as screen only	DC01 + CLIENT01 on ThinkPad; ProBook RDP client	Remote Desktop reaches CLIENT01 without joining ProBook host to the lab domain.
6. Exchange planning	Assess hardware and design	DC01 + EX01, not always CLIENT01	EX01 is a separate member server, not the DC; install only if resources allow.
7. OWA defensive validation	Mitigation/ logging only	DC01 + EX01 + browser briefly	OWA exists, mitigation checks run; no exploit reproduction.

Minimum weekly learning plan

- **Session 1:** fix/confirm DC static IP and DNS.
- **Session 2:** build Windows 10 Pro client VM.
- **Session 3:** join the domain and log in as a domain user.
- **Session 4:** create users, groups, shares and permissions.
- **Session 5:** create and test Group Policy.
- **Session 6:** practise troubleshooting with broken DNS, broken time, and wrong credentials.

4. Naming, IP plan, and VM sizing

Recommended names

Role	VM name	Computer name	Purpose
Domain Controller	DC01	DC01	AD DS, DNS, authentication, Group Policy.
Windows Client	CLIENT01	CLIENT01	Windows 10 Pro test device joined to domain.
Future Exchange	EX01	EX01	Separate member server for Exchange/OWA if hardware allows.

Internal Network IP plan

Use this when both DC01 and CLIENT01 run on the ThinkPad and communicate through a VirtualBox Internal Network called `JoshLab-Internal`.

VM	IP	Subnet	Gateway	DNS
DC01	10.10.10.10	255.255.255.0	blank	127.0.0.1
CLIENT01	10.10.10.20	255.255.255.0	blank	10.10.10.10
EX01 later	10.10.10.30	255.255.255.0	blank	10.10.10.10

Suggested VM sizing for your Lenovo

VM	RAM	CPU	Disk	Notes
DC01	2-4 GB	1-2 cores	60 GB dynamic	Server Core would be lighter, but Desktop Experience is easier for learning.
CLIENT01	2-4 GB	1-2 cores	60 GB dynamic	Windows 10 Pro 22H2 x64 is lighter than Windows 11.
EX01	8-16 GB bare minimum for lab pain	2-4 cores	120 GB+ dynamic	Do later only. Exchange is heavy and Microsoft production recommendations are much higher.

Resource rule

For your ThinkPad, the practical pattern is DC01 + CLIENT01 for AD learning, then DC01 + EX01 for Exchange learning, not all three all the time.

5. VirtualBox networking scenarios

Most AD lab failures are really DNS/network design failures. This chapter is your decision map.

Scenario A - NAT only

Use case	Works for	Problems
One VM with internet access	Installing updates, downloading tools	Other VMs and other laptops usually cannot reach the VM. `10.0.2.15` is NAT-local.

```
VirtualBox Settings > Network > Adapter 1
Attached to: NAT

Typical guest IP: 10.0.2.15
Typical gateway: 10.0.2.2
```

Good for installing Windows and updates. Bad as the final AD lab network if you need a client VM to join the domain.

Scenario B - Internal Network

Use case	Works for	Problems
Clean two-VM AD lab on one host	DC01 and CLIENT01 talk privately	The host and ProBook cannot directly remote into the guest unless you add another adapter.

```
VirtualBox Settings > Network > Adapter 1
Attached to: Internal Network
Name: JoshLab-Internal
```

This is the cleanest model for AD learning.

Scenario C - Host-only Adapter

Use case	Works for	Problems
Host needs to reach VMs	RDP/browser from ThinkPad host to VMs	Not automatically reachable from the ProBook unless routed or bridged.

```
VirtualBox Settings > Network > Adapter 1
Attached to: Host-only Adapter
Name: VirtualBox Host-Only Ethernet Adapter
```

Scenario D - Bridged Adapter

Use case	Works for	Problems
VM appears on normal home LAN	ProBook can RDP to ThinkPad-hosted client VM	School/work networks may block or monitor this. IP can change unless reserved/static.

```
VirtualBox Settings > Network > Adapter 1
Attached to: Bridged Adapter
Name: your Wi-Fi or Ethernet adapter
```

Scenario E - Dual adapter client for ProBook remote control

This is the plan you suggested, and it is good. CLIENT01 runs on the ThinkPad but is visible from the ProBook via Remote Desktop.

VM	Adapter 1	Adapter 2	Why
DC01	Internal Network: JoshLab-Internal	Optional NAT for updates only	Keeps AD private.
CLIENT01	Internal Network: JoshLab-Internal	Bridged Adapter	Uses internal adapter for domain/DNS; bridged adapter for RDP from ProBook.

DNS gotcha

On CLIENT01, the domain adapter must use DC01 as DNS. Do not point domain DNS to your router, 8.8.8.8 or 1.1.1.1.

Scenario F - Two laptops, each running a VM

Possible, but more fragile. Both VMs need bridged networking or another routable network. This is less clean than running both VMs on the ThinkPad and remoting in.

VBoxManage command examples

Run these on the Windows host in a normal Command Prompt or PowerShell, adjusting VM names and adapter names.

```
cd "C:\Program Files\Oracle\VirtualBox"

# Put DC01 on an internal network.
.\VBoxManage.exe modifyvm "DC01" --nic1 intnet --intnet1 "JoshLab-
Internal"

# Put CLIENT01 on the same internal network.
.\VBoxManage.exe modifyvm "CLIENT01" --nic1 intnet --intnet1 "JoshLab-
Internal"

# Add a bridged second adapter to CLIENT01 for RDP from the ProBook.
.\VBoxManage.exe modifyvm "CLIENT01" --nic2 bridged --bridgeadapter2
"Intel(R) Wi-Fi 6 AX201"

# Show VM network config.
.\VBoxManage.exe showvminfo "CLIENT01" | findstr /i "NIC"
```

6. Build and confirm DC01

Rename server

```
Rename-Computer -NewName "DC01" -Restart
```

Install AD DS and DNS roles

```
Install-WindowsFeature AD-Domain-Services,DNS -IncludeManagementTools
```

Create the forest

Your current lab uses `josh.lab`. Keep it simple and local to your lab.

```
Install-ADDSForest `
  -DomainName "josh.lab" `
  -DomainNetbiosName "JOSHLAB" `
  -InstallDNS `
  -Force
```

Confirm core services

```
Get-Service ADWS,KDC,Netlogon,DNS
Get-ADDomain
Get-ADForest
hostname
whoami /fqdn
```

Expected service result

```
Status   Name       DisplayName
------   ----       -----------
Running  ADWS       Active Directory Web Services
Running  DNS        DNS Server
```

```
Running   KDC         Kerberos Key Distribution Center
Running   Netlogon    Netlogon
```

What these services mean

Service	Meaning
ADWS	Lets modern AD management tools talk to Active Directory.
DNS	Holds domain records like `_ldap._tcp.josh.lab` and resolves `josh.lab`.
KDC	Kerberos authentication. Domain logon depends on this.
Netlogon	Domain logon support and secure channel support.

7. Static IP and DNS setup

Domain controllers should use stable IP addresses. Your earlier output warned that the DC had a dynamic IP. That is one of the first things to clean up.

Check current network config

```
Get-NetIPConfiguration
Get-NetAdapter
ipconfig /all
```

Internal Network static IP for DC01

```
# Adjust InterfaceAlias if your adapter is not named Ethernet.
Get-NetAdapter

New-NetIPAddress `
  -InterfaceAlias "Ethernet" `
  -IPAddress 10.10.10.10 `
  -PrefixLength 24

Set-DnsClientServerAddress `
  -InterfaceAlias "Ethernet" `
  -ServerAddresses 127.0.0.1
```

If you are temporarily using NAT only

This can work for one-VM testing, but is not ideal for client domain join.

```
# NAT-style example only.
New-NetIPAddress `
  -InterfaceAlias "Ethernet" `
  -IPAddress 10.0.2.15 `
  -PrefixLength 24 `
  -DefaultGateway 10.0.2.2

Set-DnsClientServerAddress `
```

```
  -InterfaceAlias "Ethernet" `
  -ServerAddresses 127.0.0.1
```

Confirm DNS records

```
nslookup josh.lab 127.0.0.1
nslookup _ldap._tcp.dc._msdcs.josh.lab 127.0.0.1
Resolve-DnsName josh.lab
Resolve-DnsName _ldap._tcp.dc._msdcs.josh.lab -Type SRV
```

> **Rule to remember**
>
> Domain clients find domain controllers through DNS SRV records. If DNS is wrong, AD looks broken even when AD itself is fine.

8. DC health checks and how to read them

Core diagnostics

```
dcdiag /c /v
dcdiag /q
dcdiag /test:dns /v
dcdiag /test:dfsrevent /v
repadmin /replsummary
repadmin /showrepl
Get-SmbShare SYSVOL,NETLOGON
Get-Service DFSR
w32tm /query /status
```

How to interpret common results

Output	Meaning	Action
`passed test Connectivity`	LDAP/RPC to the DC works.	Good.
`passed test Advertising`	The DC advertises itself as DC, LDAP, KDC, GC/time server.	Good.
`passed test DNS`	DNS records exist and dynamic updates work.	Good.
`failed test DFSREvent` shortly after build	Often old event log errors from setup/startup.	Confirm DFSR is running and SYSVOL/NETLOGON are online; re-test later.
`failed test SystemLog` with unexpected shutdown	The VM was powered off roughly or rebooted during setup.	Shut down cleanly; ignore if not recurring.
`Warning IP address is dynamic`	DC uses DHCP.	Set a static IP.
`time service ... offset`	VM time jumped.	Fix time sync.

DFSR specific checks

```
Get-Service DFSR
dfsrdiag pollad
dcdiag /test:dfsrevent /v
Get-SmbShare SYSVOL,NETLOGON
```

Why DFSREvent can still fail

`dcdiag` checks recent event logs. It can report old DFSR warnings/errors for up to 24 hours after SYSVOL was shared. If `DFSR` is running, `dfsrdiag pollad` succeeds, and `SYSVOL` / `NETLOGON` are online, the domain may be fine even while the event test still complains.

9. Time service and Kerberos

AD authentication relies heavily on time. Big time jumps can cause Kerberos failures, domain join failures and strange login issues.

Check time

```
w32tm /query /status
w32tm /query /configuration
Get-Date
```

Force a resync

```
w32tm /resync /force
```

Configure the PDC emulator to use an external time source

For a small lab, DC01 is the forest root PDC emulator. During a connected/update window, you can set it as reliable and sync to an external source.

```
w32tm /config /manualpeerlist:"time.windows.com" /syncfromflags:manual /
reliable:yes /update
Restart-Service w32time
w32tm /resync /force
w32tm /query /status
```

Client time check after domain join

```
w32tm /query /source
w32tm /query /status
nltest /dsgetdc:josh.lab
```

10. Create users, groups, and OUs

Create a simple OU layout

```
New-ADOrganizationalUnit -Name "Lab" -Path "DC=josh,DC=lab"
New-ADOrganizationalUnit -Name "Users" -Path "OU=Lab,DC=josh,DC=lab"
New-ADOrganizationalUnit -Name "Computers" -Path "OU=Lab,DC=josh,DC=lab"
New-ADOrganizationalUnit -Name "Groups" -Path "OU=Lab,DC=josh,DC=lab"
```

Create a test user

```
New-ADUser `
  -Name "Test User" `
  -SamAccountName "testuser" `
  -UserPrincipalName "testuser@josh.lab" `
  -Path "OU=Users,OU=Lab,DC=josh,DC=lab" `
  -AccountPassword (Read-Host -AsSecureString "Enter password") `
  -Enabled $true
```

Common account commands

```
Get-ADUser testuser -Properties *
Set-ADAccountPassword testuser -Reset -NewPassword (Read-Host
-AsSecureString "New password")
Unlock-ADAccount testuser
Disable-ADAccount testuser
Enable-ADAccount testuser
Set-ADUser testuser -ChangePasswordAtLogon $false
```

Create groups and add the user

```
New-ADGroup `
  -Name "Lab Share Users" `
  -SamAccountName "LabShareUsers" `
  -GroupScope Global `
  -GroupCategory Security `
```

```
  -Path "OU=Groups,OU=Lab,DC=josh,DC=lab"

Add-ADGroupMember -Identity "Lab Share Users" -Members testuser
Get-ADGroupMember "Lab Share Users"
```

11. Build CLIENT01 with Windows 10 Pro

Edition recommendation

Use **Windows 10 Pro 22H2 x64** for the client VM. The ISO is free to download from Microsoft, but Windows itself still requires a valid licence for long-term/activated use. During install, choose **I do not have a product key**, then choose **Windows 10 Pro**. Do not use Windows Home, because Home cannot join an on-prem Active Directory domain.

Download flow

1. Open Microsoft Windows 10 download page.
2. Download the Media Creation Tool.
3. Choose **Create installation media for another PC**.
4. Choose English, Windows 10, 64-bit x64.
5. Choose ISO file.
6. Attach the ISO to the CLIENT01 VM in VirtualBox.

VirtualBox VM settings

Setting	Value
Name	CLIENT01
Type/version	Microsoft Windows / Windows 10 64-bit
RAM	4096 MB if possible, 2048 MB minimum
CPU	2 cores if possible
Disk	60 GB dynamic VDI
Network Adapter 1	Internal Network: `JoshLab-Internal`
Network Adapter 2 optional	Bridged Adapter if you want RDP from the ProBook

After install, rename the client

```
Rename-Computer -NewName "CLIENT01" -Restart
```

12. Configure CLIENT01 networking

Internal adapter only

```
Get-NetAdapter

New-NetIPAddress `
  -InterfaceAlias "Ethernet" `
  -IPAddress 10.10.10.20 `
  -PrefixLength 24

Set-DnsClientServerAddress `
  -InterfaceAlias "Ethernet" `
  -ServerAddresses 10.10.10.10

ipconfig /all
```

Dual adapter client: Internal + Bridged

If CLIENT01 has one internal domain adapter and one bridged adapter for RDP, be careful with DNS. The internal/domain adapter should use the DC for DNS. The bridged adapter can use your home router for internet if needed, but do not let it become the preferred DNS path for domain lookup.

```
# Example: set DNS on internal/domain adapter.
Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses
10.10.10.10

# Example: if bridged adapter is Ethernet 2, leave DNS automatic or set
router as needed.
Get-DnsClientServerAddress
Get-NetIPConfiguration
```

Connectivity tests before joining the domain

```
ping 10.10.10.10
nslookup josh.lab 10.10.10.10
Resolve-DnsName josh.lab -Server 10.10.10.10
Test-NetConnection 10.10.10.10 -Port 53
```

```
Test-NetConnection 10.10.10.10 -Port 88
Test-NetConnection 10.10.10.10 -Port 389
Test-NetConnection 10.10.10.10 -Port 445
nltest /dsgetdc:josh.lab
```

Ports to understand

Port	Service	Why it matters
53	DNS	Client finds the domain and DC records.
88	Kerberos	Domain authentication.
389	LDAP	Directory queries.
445	SMB	SYSVOL, NETLOGON and file shares.
3389	RDP	Remote control if enabled.

13. Join CLIENT01 to josh.lab

PowerShell method

```
Add-Computer -DomainName josh.lab -Credential JOSHLAB\\Administrator
-Restart
```

GUI method

1. Open **Control Panel**.
2. Go to **System and Security** then **System**.
3. Select **Change settings**.
4. Computer Name tab - **Change**.
5. Member of - select **Domain**.
6. Enter `josh.lab`.
7. Use `JOSHLAB\Administrator` credentials.
8. Restart when prompted.

After reboot

```
# Log on as:
JOSHLAB\testuser

# Then run:
whoami
whoami /fqdn
gpresult /r
nltest /dsgetdc:josh.lab
Test-ComputerSecureChannel -Verbose
```

Move computer object into your lab OU

```
Get-ADComputer CLIENT01
Move-ADObject `
  -Identity (Get-ADComputer CLIENT01).DistinguishedName `
  -TargetPath "OU=Computers,OU=Lab,DC=josh,DC=lab"
```

Common join failures

Error / symptom	Likely cause	Fix
Domain cannot be contacted	Client DNS is not pointing to DC01.	Set client DNS to `10.10.10.10` and retry `nltest /dsgetdc:josh.lab`.
Credential prompt rejects credentials	Wrong domain prefix or password.	Use `JOSHLAB\Administrator` or `administrator@josh.lab`.
Time/Kerberos errors	Client and DC clocks differ too much.	Fix `w32tm` and resync.
Ping works but join fails	DNS SRV records or firewall issue.	Test ports 53, 88, 389, 445 and run `dcdiag /test:dns /v`.

14. Remote into CLIENT01 from the ProBook

This lets the ProBook act as a screen/keyboard while the VM still runs on the ThinkPad.

Enable RDP on CLIENT01

```
Set-ItemProperty `
  -Path "HKLM:\System\CurrentControlSet\Control\Terminal Server" `
  -Name "fDenyTSConnections" `
  -Value 0

Enable-NetFirewallRule -DisplayGroup "Remote Desktop"

# Optional: allow your domain test user to use RDP.
Add-LocalGroupMember -Group "Remote Desktop Users" -Member
"JOSHLAB\testuser"
```

Find CLIENT01 bridged IP

```
ipconfig
Get-NetIPConfiguration
```

From the ProBook

```
mstsc
```

Computer: the bridged IP address of CLIENT01. Username: `JOSHLAB\testuser`.

> **Do not join the ProBook host**
> The ProBook should remain outside the lab domain. It only opens Remote Desktop to CLIENT01.

If RDP fails

```
# From ProBook PowerShell, replace IP with CLIENT01 bridged IP.
Test-NetConnection 192.168.0.50 -Port 3389
ping 192.168.0.50
```

15. First practical AD tests

Test 1 - domain user login

```
# On CLIENT01 after domain login
whoami
whoami /groups
whoami /fqdn
Test-ComputerSecureChannel -Verbose
```

Test 2 - create a domain share

```
# On DC01
New-Item -ItemType Directory -Path C:\LabShare -Force
New-SmbShare -Name LabShare -Path C:\LabShare -ChangeAccess "JOSHLAB\Lab
Share Users"

# NTFS permissions - grant modify to group.
$acl = Get-Acl C:\LabShare
$rule = New-Object
System.Security.AccessControl.FileSystemAccessRule("JOSHLAB\Lab Share
Users","Modify","ContainerInherit,ObjectInherit","None","Allow")
$acl.AddAccessRule($rule)
Set-Acl C:\LabShare $acl

# Create a test file.
"Hello from the domain" | Out-File C:\LabShare\hello.txt
```

Test share from CLIENT01

```
dir \\DC01\LabShare
New-PSDrive -Name L -PSProvider FileSystem -Root \\DC01\LabShare
-Persist
notepad L:\hello.txt
```

Test 3 - Group Policy smoke test

```
# On DC01
New-GPO -Name "Lab - First Test Policy"
New-GPLink -Name "Lab - First Test Policy" -Target
"OU=Computers,OU=Lab,DC=josh,DC=lab"

# On CLIENT01
gpupdate /force
gpresult /r
gpresult /h C:\Users\Public\Desktop\gpresult.html
```

For a visible first policy, use the Group Policy Management GUI and configure something harmless, such as a desktop wallpaper or control panel restriction for the lab OU.

16. Group Policy learning path

Recommended starter policies

Policy	Why it teaches well	Risk
Map a drive	Teaches users, groups, shares and GPP.	Low.
Set wallpaper	Very visible proof that GPO applies.	Low.
Disable Control Panel for test user	Teaches user policy targeting.	Medium - can confuse you if applied too broadly.
Add test local admin group	Teaches restricted groups/local users and groups.	Medium - understand before using.
Windows Firewall rule for RDP	Teaches endpoint policy.	Medium.

Useful GPO commands

```
Import-Module GroupPolicy
Get-GPO -All
New-GPO -Name "Lab - Test"
New-GPLink -Name "Lab - Test" -Target
"OU=Computers,OU=Lab,DC=josh,DC=lab"
Get-GPInheritance -Target "OU=Computers,OU=Lab,DC=josh,DC=lab"
gpupdate /force
gpresult /r
gpresult /h C:\Temp\gpresult.html
rsop.msc
```

GPO troubleshooting model

1. Is the computer/user object in the OU where the GPO is linked?
2. Is the GPO linked and enabled?
3. Is security filtering excluding the object?
4. Does the setting apply to Computer Configuration or User Configuration?
5. Has the client refreshed policy?
6. Does `gpresult` show the policy as applied or denied?

17. Troubleshooting decision trees

Domain join fails

```
# On CLIENT01
ipconfig /all
nslookup josh.lab 10.10.10.10
Resolve-DnsName _ldap._tcp.dc._msdcs.josh.lab -Type SRV -Server
10.10.10.10
Test-NetConnection 10.10.10.10 -Port 53
Test-NetConnection 10.10.10.10 -Port 88
Test-NetConnection 10.10.10.10 -Port 389
Test-NetConnection 10.10.10.10 -Port 445
w32tm /query /status
nltest /dsgetdc:josh.lab
```

Client joined but cannot log in

```
# On CLIENT01 local admin session
Test-ComputerSecureChannel -Verbose
nltest /sc_query:josh.lab
klist
w32tm /query /status

# On DC01
Get-ADUser testuser -Properties LockedOut,Enabled,BadLogonCount
Search-ADAccount -LockedOut
Unlock-ADAccount testuser
```

SYSVOL or NETLOGON missing

```
# On DC01
Get-SmbShare SYSVOL,NETLOGON
Get-Service DFSR
dfsrdiag pollad
dcdiag /test:sysvolcheck /v
```

```
dcdiag /test:netlogons /v
dcdiag /test:dfsrevent /v
```

DNS looks broken

```
# On DC01
Get-DnsServerZone
Get-DnsServerResourceRecord -ZoneName josh.lab
ipconfig /registerdns
Restart-Service Netlogon
Restart-Service DNS
dcdiag /test:dns /v

# On CLIENT01
ipconfig /flushdns
ipconfig /registerdns
Resolve-DnsName josh.lab -Server 10.10.10.10
```

Time looks broken

```
# On DC01
w32tm /query /status
w32tm /resync /force

# On CLIENT01
w32tm /query /source
w32tm /resync /force
```

RDP does not work

```
# On CLIENT01
Get-NetFirewallRule -DisplayGroup "Remote Desktop" | Select
DisplayName,Enabled
Enable-NetFirewallRule -DisplayGroup "Remote Desktop"
Get-ItemProperty "HKLM:\System\CurrentControlSet\Control\Terminal
Server" -Name fDenyTSConnections
```

```
# On ProBook
Test-NetConnection <CLIENT01-BRIDGED-IP> -Port 3389
```

18. Exchange and OWA: what your lab does and does not have

Your current Exchange conclusion

Your check returned `vmickvpexchange`, which is **Hyper-V Data Exchange Service**, not Microsoft Exchange Server. `Test-Path "C:\Program Files\Microsoft\Exchange Server\V15"` returned `False`. Therefore Exchange/OWA is not installed and the OWA CVE is not applicable to the current VM.

Commands to confirm Exchange presence

```
Get-Service *Exchange*
Test-Path "C:\Program Files\Microsoft\Exchange Server\V15"
Get-Command ExSetup.exe -ErrorAction SilentlyContinue
Get-Website | Where-Object Name -like "*Exchange*"
Get-WebApplication | Where-Object Path -like "*owa*"
```

Do not install Exchange on DC01

For clean learning, keep Exchange separate. AD and Exchange are both heavy and security-sensitive. In production-style design, Exchange should be on a member server. Once Exchange is installed, changing the server between member server and domain controller roles is not supported.

Future Exchange lab shape

VM	Role	IP	When to run
DC01	Domain Controller and DNS	10.10.10.10	Always needed for domain.
EX01	Exchange member server	10.10.10.30	Run with DC01 only when practising Exchange.
CLIENT01	Browser/client test	10.10.10.20	Run briefly for OWA browser testing or use host browser if reachable.

Resource-limited operating pattern

```
Normal AD practice:       DC01 + CLIENT01
Exchange install/practice: DC01 + EX01
OWA smoke test:            DC01 + EX01 + CLIENT01 briefly, or DC01 +
EX01 + host browser
Do not expect:             DC01 + CLIENT01 + EX01 all day on a small
ThinkPad
```

19. Safe Exchange planning guide

Before you even try Exchange

- Finish AD client domain join first.
- Take a snapshot of DC01.
- Confirm DC01 has static IP and stable DNS.
- Confirm the ThinkPad has enough spare RAM and disk.
- Expect Exchange to be slow in a small lab.

High-level build order

```
1. Build DC01 and josh.lab.
2. Build EX01 as a separate Windows Server VM.
3. Set EX01 DNS to DC01.
4. Join EX01 to josh.lab.
5. Install Exchange prerequisites using the current Microsoft
documentation.
6. Prepare AD using Exchange Setup if required.
7. Install the Exchange Mailbox role.
8. Create test mailboxes.
9. Test OWA login and ordinary mail flow.
10. Run defensive mitigation/health checks.
```

EX01 basic member-server prep

```
Rename-Computer -NewName "EX01" -Restart

New-NetIPAddress `
  -InterfaceAlias "Ethernet" `
  -IPAddress 10.10.10.30 `
  -PrefixLength 24

Set-DnsClientServerAddress `
  -InterfaceAlias "Ethernet" `
  -ServerAddresses 10.10.10.10
```

```
Add-Computer -DomainName josh.lab -Credential JOSHLAB\\Administrator
-Restart
```

Exchange setup reminder

Use the current Microsoft Exchange prerequisites and setup documentation for the exact version you install. Do not rely on random blog scripts. Exchange version, Windows Server version, .NET requirements and Visual C++ dependencies can change by build.

20. OWA / CVE defensive validation only

> **Boundary**
>
> This chapter intentionally avoids exploit reproduction. It is for checking whether Exchange/OWA exists, whether mitigations are present, and whether logs/evidence are being collected.

Check build and services

```
# Exchange Management Shell on EX01
Get-Command ExSetup.exe | ForEach-Object { $_.FileVersionInfo }
Get-Service *Exchange*
Get-ExchangeServer | Format-List
Name,Edition,AdminDisplayVersion,MitigationsEnabled,MitigationsApplied,Mitigations
```

Check Exchange Emergency Mitigation service

```
Get-Service MSExchangeMitigation
Get-OrganizationConfig | Format-List MitigationsEnabled
Get-ExchangeServer | Format-List
Name,MitigationsEnabled,MitigationsApplied,MitigationsBlocked

cd "$env:ExchangeInstallPath\Scripts"
.\Test-MitigationServiceConnectivity.ps1
```

Check OWA / IIS rewrite rules defensively

```
Import-Module WebAdministration

Get-WebConfiguration `
  -PSPath 'IIS:\Sites\Default Web Site\owa' `
  -Filter 'system.webServer/rewrite/outboundRules/rule' |
  Select-Object name,enabled,preCondition |
  Format-Table -AutoSize
```

Browser-visible mitigation check

1. From a lab client, open OWA normally.
2. Open Developer Tools.
3. Network tab - reload OWA.
4. Select the main OWA HTML response.
5. Inspect Response Headers.
6. Look for the expected defensive Content Security Policy header from Microsoft mitigation guidance.

Safe OWA smoke tests

Test	Expected result
Login to OWA with test user	Login succeeds.
Read plain-text email	Message opens normally.
Read normal formatted email	Bold, bullet list and harmless link render normally.
Compose and send mail	Recipient receives message.
Reply and forward ordinary messages	No UI breakage.
Calendar opens	Basic calendar UI works.
Check mitigation/header state	Header/rule is present if applicable.

Do not send malformed HTML, script tags, event handlers, or CVE-specific payloads through OWA.

21. Evidence collection and reporting

Create an evidence folder

```
New-Item -ItemType Directory -Path C:\LabEvidence -Force
```

Capture AD health

```
hostname | Out-File C:\LabEvidence\01-hostname.txt
Get-NetIPConfiguration | Out-File C:\LabEvidence\02-network.txt
Get-Service ADWS,KDC,Netlogon,DNS,DFSR | Out-File C:\LabEvidence\03-
services.txt
dcdiag /q | Out-File C:\LabEvidence\04-dcdiag-quick.txt
dcdiag /test:dns /v | Out-File C:\LabEvidence\05-dcdiag-dns.txt
Get-SmbShare SYSVOL,NETLOGON | Format-List * | Out-File C:
\LabEvidence\06-sysvol-netlogon.txt
w32tm /query /status | Out-File C:\LabEvidence\07-time.txt
```

Capture client domain status

```
whoami /all | Out-File C:\LabEvidence\client-whoami.txt
gpresult /r | Out-File C:\LabEvidence\client-gpresult.txt
gpresult /h C:\LabEvidence\client-gpresult.html
Test-ComputerSecureChannel -Verbose | Out-File C:\LabEvidence\client-
secure-channel.txt
nltest /dsgetdc:josh.lab | Out-File C:\LabEvidence\client-dsgetdc.txt
```

Capture Exchange defensive status later

```
Get-Command ExSetup.exe | ForEach-Object { $_.FileVersionInfo } | Out-
File C:\LabEvidence\exchange-build.txt
Get-Service MSExchangeMitigation | Format-List * | Out-File C:
\LabEvidence\exchange-em-service.txt
Get-ExchangeServer | Format-List
```

```
Name,Edition,AdminDisplayVersion,MitigationsEnabled,MitigationsApplied,Mitigations
| Out-File C:\LabEvidence\exchange-mitigations.txt
```

PowerShell transcript wrapper

```
Start-Transcript C:\LabEvidence\session-transcript.txt
# Run commands here.
Stop-Transcript
```

22. Snapshot and rollback strategy

When to snapshot

Snapshot name	When
01-Clean-OS	After Windows install, before roles.
02-Pre-ADDS	Before promoting DC01.
03-Post-AD-Healthy	After AD, DNS, SYSVOL and time are working.
04-Pre-Client-Join	Before joining CLIENT01 to the domain.
05-Post-Client-Join	After domain login works.
06-Pre-GPO-Experiments	Before testing policies that could lock you out.
07-Pre-Exchange	Before installing Exchange prerequisites or schema changes.

Snapshot discipline

- Shut down cleanly before snapshotting if possible.
- Do not keep long chains of snapshots forever; they can slow VMs and eat disk.
- Label snapshots with purpose, not just dates.
- Take evidence before rolling back so you still learn from the failure.

Clean shutdown commands

```
shutdown /s /t 0
Restart-Computer
Stop-Computer
```

23. Security hardening learning path

Do not harden everything immediately. First prove the lab works. Then harden one setting at a time and observe the effect.

Topics to learn after basics

Topic	Why it matters	Starter command/check
LDAP signing	Prevents unsafe unsigned LDAP binds.	Review Directory Service event warnings first.
LDAPS	Encrypts LDAP traffic.	Requires certificates.
Channel binding	Improves LDAPS bind security.	Learn after LDAPS works.
NTLM reduction	Reduces older authentication dependency.	Audit before enforcing.
Least privilege	Avoid using Domain Admin for daily tasks.	Create a normal admin group and delegate.
LAPS / Windows LAPS	Controls local admin passwords.	Learn after GPO basics.
Firewall profiles	Understand Domain/Private/Public.	`Get-NetFirewallProfile`

Useful security visibility commands

```
Get-EventLog -LogName Security -Newest 20
Get-WinEvent -LogName "Directory Service" -MaxEvents 20
Get-ADDefaultDomainPasswordPolicy
Get-ADUser -Filter * -Properties PasswordLastSet,Enabled | Select
Name,Enabled,PasswordLastSet
Get-ADGroupMember "Domain Admins"
```

24. Command library

Server identity

```
hostname
Rename-Computer -NewName "DC01" -Restart
Get-ComputerInfo | Select
CsName,WindowsProductName,WindowsVersion,OsHardwareAbstractionLayer
```

Networking

```
Get-NetAdapter
Get-NetIPConfiguration
Get-DnsClientServerAddress
New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 10.10.10.10
-PrefixLength 24
Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses
127.0.0.1
ipconfig /all
ipconfig /flushdns
ipconfig /registerdns
route print
Test-NetConnection 10.10.10.10 -Port 389
```

AD DS

```
Install-WindowsFeature AD-Domain-Services,DNS -IncludeManagementTools
Install-ADDSForest -DomainName "josh.lab" -DomainNetbiosName "JOSHLAB"
-InstallDNS -Force
Get-ADDomain
Get-ADForest
Get-ADDomainController
Get-ADDomainController -Discover
```

Health

```
dcdiag /q
dcdiag /c /v
dcdiag /test:dns /v
dcdiag /test:dfsrevent /v
repadmin /replsummary
repadmin /showrepl
Get-Service ADWS,KDC,Netlogon,DNS,DFSR
Get-SmbShare SYSVOL,NETLOGON
```

Users and groups

```
New-ADUser -Name "Test User" -SamAccountName testuser -UserPrincipalName
testuser@josh.lab -AccountPassword (Read-Host -AsSecureString
"Password") -Enabled $true
Get-ADUser testuser -Properties *
Set-ADAccountPassword testuser -Reset -NewPassword (Read-Host
-AsSecureString "New password")
Unlock-ADAccount testuser
New-ADGroup -Name "Lab Share Users" -GroupScope Global -GroupCategory
Security
Add-ADGroupMember "Lab Share Users" testuser
Get-ADGroupMember "Lab Share Users"
```

Domain join and trust

```
Add-Computer -DomainName josh.lab -Credential JOSHLAB\\Administrator
-Restart
Remove-Computer -UnjoinDomainCredential JOSHLAB\\Administrator
-WorkgroupName WORKGROUP -Restart
Test-ComputerSecureChannel -Verbose
Test-ComputerSecureChannel -Repair -Credential JOSHLAB\\Administrator
nltest /dsgetdc:josh.lab
nltest /sc_query:josh.lab
```

Group Policy

```
Import-Module GroupPolicy
Get-GPO -All
New-GPO -Name "Lab - Test"
New-GPLink -Name "Lab - Test" -Target
"OU=Computers,OU=Lab,DC=josh,DC=lab"
gpupdate /force
gpresult /r
gpresult /h C:\Temp\gpresult.html
rsop.msc
```

Remote Desktop and WinRM

```
Set-ItemProperty -Path "HKLM:\System\CurrentControlSet\Control\Terminal
Server" -Name "fDenyTSConnections" -Value 0
Enable-NetFirewallRule -DisplayGroup "Remote Desktop"
Test-NetConnection <IP> -Port 3389
Enable-PSRemoting -Force
Test-WSMan <ComputerName>
Enter-PSSession <ComputerName> -Credential JOSHLAB\\Administrator
```

Exchange defensive checks later

```
Get-Service *Exchange*
Test-Path "C:\Program Files\Microsoft\Exchange Server\V15"
Get-Command ExSetup.exe -ErrorAction SilentlyContinue
Get-Service MSExchangeMitigation
Get-ExchangeServer | Format-List
Name,MitigationsEnabled,MitigationsApplied,MitigationsBlocked
```

25. Glossary

Term	Meaning
AD DS	Active Directory Domain Services. The identity system for users, computers and groups.
DC	Domain Controller. Server that hosts AD DS.
DNS	Name resolution. AD depends on DNS heavily.
KDC	Kerberos Key Distribution Center. Handles Kerberos tickets for domain auth.
SYSVOL	Domain share containing Group Policy and scripts.
NETLOGON	Domain logon share.
DFSR	DFS Replication. Replicates SYSVOL between domain controllers.
OU	Organizational Unit. Container for users/computers/groups and GPO links.
GPO	Group Policy Object. Settings applied to users or computers.
SPN	Service Principal Name. Used by Kerberos to identify services.
RDP	Remote Desktop Protocol.
OWA	Outlook Web Access / Outlook on the web for Exchange Server.
Exchange member server	A domain-joined server running Exchange, separate from the DC.
NAT	VirtualBox network mode giving guest outbound internet but isolating inbound access.
Bridged	VirtualBox network mode where the VM appears on the same LAN as the host.
Internal Network	VirtualBox private VM-to-VM network on the same host.
Host-only	VirtualBox network between host and guests.

26. Final recommended next actions

Do next

1. Create CLIENT01 as Windows 10 Pro 22H2 x64.
2. Put DC01 and CLIENT01 on `JoshLab-Internal`.
3. Set DC01 static IP to `10.10.10.10` and DNS to `127.0.0.1`.
4. Set CLIENT01 static IP to `10.10.10.20` and DNS to `10.10.10.10`.
5. Run connectivity tests from CLIENT01.
6. Join CLIENT01 to `josh.lab`.
7. Log in as `JOSHLAB\testuser`.
8. Test a share and a simple GPO.
9. Snapshot the clean working state.

Do not do yet

- Do not install Exchange on DC01 to save RAM.
- Do not try OWA CVE testing until Exchange/OWA actually exists.
- Do not join the DCS ProBook host to your lab domain.
- Do not run exploit payloads or crafted OWA exploit emails.

One-line success definition

```
CLIENT01 joins josh.lab, testuser logs in, gpresult shows domain policy,
and \\DC01\LabShare opens successfully.
```

References

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   https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/deploy/install-active-directory-domain-services--level-100-
5. Microsoft Learn - Add-Computer PowerShell cmdlet
   https://learn.microsoft.com/en-us/powershell/module/microsoft.powershell.management/add-computer
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Generated as a lab reference. Use official vendor documentation for production changes and current Exchange prerequisites.